

## KUTAISI INTL, Georgia

WMO: 371775

Lat: 42.177N

Lon: 42.483E

Elev: 68

StdP: 100.51

Time Zone: 4.00 (E04)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -2.2       | -1.0 | -7.2                          | 2.1 | 5.2  | -6.0 | 2.3 | 5.6  | 17.3                  | 9.3  | 15.5 | 9.5  | 3.4                   | 90   | 0.479 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 10.4                   | 35.0            | 23.2 | 33.0 | 23.1 | 31.1 | 22.5 | 25.7                | 31.1 | 24.9 | 29.8 | 24.1 | 28.6 | 7.7                  | 90   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 24.1                            | 19.2 | 28.5 | 23.2 | 18.1 | 27.5 | 22.9 | 17.8 | 27.1 | 79.4          | 31.0 | 76.0 | 30.0 | 73.0 | 29.1 | 29.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.4          | 14.4 | 12.1 | DB                         | -5.2 | 38.4               | 2.7 | 1.9                                                | -7.2 | 39.8       | -8.7 | 40.9       | -10.2 | 41.9       | -12.2 | 43.3 | (4) |
| (5)               |      |      | WB                         | -5.4 | 27.3               | 2.8 | 1.4                                                | -7.5 | 28.3       | -9.1 | 29.1       | -10.7 | 29.9       | -12.8 | 30.9 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 16.0        | 6.8   | 7.7   | 10.3  | 15.0  | 19.1  | 22.8  | 24.7  | 25.5  | 22.1  | 17.2  | 11.9  | 8.1   | (6)  |  |
|                                                                        | DBStd       | 7.51        | 3.40  | 4.16  | 4.40  | 3.95  | 3.43  | 2.56  | 2.53  | 2.52  | 3.60  | 3.49  | 3.95  | 4.16  | (7)  |  |
|                                                                        | HDD10.0     | 360         | 108   | 86    | 51    | 3     | 0     | 0     | 0     | 0     | 0     | 1     | 26    | 86    | (8)  |  |
|                                                                        | HDD18.3     | 1638        | 357   | 299   | 249   | 113   | 33    | 1     | 0     | 0     | 12    | 64    | 192   | 317   | (9)  |  |
|                                                                        | CDD10.0     | 2544        | 9     | 21    | 61    | 154   | 282   | 384   | 456   | 480   | 362   | 223   | 84    | 27    | (10) |  |
|                                                                        | CDD18.3     | 779         | 0     | 0     | 1     | 14    | 57    | 136   | 198   | 222   | 123   | 28    | 1     | 0     | (11) |  |
|                                                                        | CDH23.3     | 6622        | 0     | 1     | 9     | 187   | 518   | 1078  | 1648  | 2007  | 969   | 191   | 13    | 1     | (12) |  |
|                                                                        | CDH26.7     | 2444        | 0     | 0     | 0     | 38    | 165   | 368   | 636   | 849   | 353   | 33    | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 4.3         | 4.7   | 5.7   | 5.3   | 4.9   | 3.7   | 3.5   | 3.2   | 3.4   | 3.7   | 4.1   | 4.6   | 4.7   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1441        | 137   | 128   | 98    | 83    | 64    | 111   | 84    | 113   | 132   | 152   | 152   | 189   | (15) |  |
|                                                                        | PrecMax     | 1909        | 307   | 300   | 200   | 250   | 110   | 268   | 247   | 350   | 284   | 309   | 454   | 405   | (16) |  |
|                                                                        | PrecMin     | 1066        | 27    | 13    | 38    | 20    | 14    | 46    | 15    | 9     | 34    | 15    | 16    | 44    | (17) |  |
|                                                                        | PrecStd     | 217         | 74    | 70    | 44    | 43    | 25    | 49    | 55    | 62    | 61    | 81    | 93    | 80    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 17.1  | 21.9  | 24.8  | 30.1  | 33.0  | 35.0  | 38.0  | 38.1  | 36.1  | 29.9  | 25.0  | 21.2  | (19) |  |
|                                                                        |             | MCWB        | 8.6   | 11.6  | 13.1  | 18.4  | 20.7  | 22.3  | 23.7  | 21.3  | 20.6  | 18.9  | 14.5  | 13.0  | (20) |  |
|                                                                        | 2%          | DB          | 15.2  | 19.0  | 22.1  | 27.2  | 30.9  | 32.1  | 34.8  | 34.9  | 32.8  | 27.2  | 22.2  | 18.1  | (21) |  |
|                                                                        |             | MCWB        | 8.3   | 9.9   | 12.4  | 16.4  | 19.7  | 22.6  | 23.5  | 23.9  | 20.8  | 18.3  | 13.7  | 10.4  | (22) |  |
|                                                                        | 5%          | DB          | 14.0  | 16.8  | 20.0  | 25.2  | 28.1  | 30.2  | 32.0  | 32.9  | 30.1  | 25.2  | 20.2  | 16.1  | (23) |  |
|                                                                        |             | MCWB        | 7.5   | 9.7   | 11.7  | 15.7  | 19.2  | 22.0  | 24.1  | 24.1  | 20.9  | 17.2  | 12.8  | 9.6   | (24) |  |
|                                                                        | 10%         | DB          | 12.2  | 14.1  | 17.9  | 23.1  | 26.1  | 28.9  | 30.0  | 31.1  | 28.2  | 23.8  | 18.2  | 14.1  | (25) |  |
|                                                                        |             | MCWB        | 6.9   | 8.5   | 10.8  | 14.5  | 18.5  | 21.5  | 23.8  | 23.5  | 20.7  | 17.3  | 12.2  | 9.0   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 10.6  | 13.1  | 14.3  | 19.1  | 22.4  | 25.3  | 26.9  | 27.2  | 24.2  | 20.7  | 16.5  | 13.7  | (27) |  |
|                                                                        |             | MCDB        | 14.3  | 19.6  | 21.4  | 28.2  | 29.2  | 30.3  | 31.5  | 32.8  | 30.2  | 26.1  | 21.1  | 18.3  | (28) |  |
|                                                                        | 2%          | WB          | 9.5   | 11.1  | 13.0  | 17.5  | 21.0  | 24.2  | 25.8  | 25.8  | 23.0  | 19.6  | 15.1  | 12.0  | (29) |  |
|                                                                        |             | MCDB        | 12.4  | 16.4  | 20.2  | 25.8  | 27.8  | 28.8  | 31.0  | 31.5  | 28.2  | 24.6  | 18.9  | 16.3  | (30) |  |
|                                                                        | 5%          | WB          | 8.6   | 10.1  | 12.2  | 16.3  | 20.0  | 23.3  | 24.9  | 25.1  | 22.3  | 18.6  | 14.1  | 10.6  | (31) |  |
|                                                                        |             | MCDB        | 11.7  | 14.4  | 18.6  | 23.8  | 26.6  | 27.9  | 29.7  | 30.2  | 27.4  | 23.3  | 19.0  | 14.4  | (32) |  |
|                                                                        | 10%         | WB          | 7.7   | 9.1   | 11.2  | 15.1  | 19.0  | 22.4  | 24.0  | 24.3  | 21.6  | 17.7  | 13.1  | 9.5   | (33) |  |
|                                                                        |             | MCDB        | 10.8  | 13.0  | 16.8  | 21.9  | 25.1  | 27.1  | 28.4  | 28.9  | 26.4  | 22.2  | 17.1  | 13.1  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.7   | 8.3   | 9.1   | 11.7  | 10.9  | 10.5  | 9.4   | 10.4  | 10.6  | 10.2  | 9.4   | 8.2   | (35) |  |
|                                                                        |             | MCDBR       | 10.3  | 12.7  | 13.3  | 16.2  | 15.1  | 13.3  | 12.8  | 13.2  | 13.8  | 13.2  | 13.6  | 11.1  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.8   | 6.3   | 6.3   | 7.9   | 7.4   | 6.0   | 5.7   | 5.6   | 6.0   | 6.4   | 7.1   | 6.5   | (37) |  |
|                                                                        |             | MCWBR       | 8.3   | 10.7  | 11.4  | 14.7  | 13.5  | 11.7  | 10.6  | 11.2  | 12.0  | 11.4  | 11.6  | 9.8   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.341 | 0.376 | 0.433 | 0.486 | 0.452 | 0.438 | 0.467 | 0.471 | 0.413 | 0.403 | 0.357 | 0.334 | (40) |  |
|                                                                        |             | taud        | 2.273 | 2.153 | 2.058 | 1.995 | 2.180 | 2.288 | 2.230 | 2.214 | 2.359 | 2.322 | 2.398 | 2.343 | (41) |  |
|                                                                        | Ebn at Noon |             | 781   | 806   | 800   | 787   | 828   | 840   | 810   | 793   | 818   | 773   | 764   | 758   | (42) |  |
|                                                                        |             | Edh at Noon | 90    | 119   | 148   | 170   | 145   | 131   | 137   | 135   | 109   | 101   | 81    | 78    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.66  | 2.50  | 3.34  | 4.50  | 5.31  | 5.88  | 5.80  | 5.44  | 4.39  | 2.97  | 2.00  | 1.46  | (44) |  |
|                                                                        | RadStd      |             | 0.23  | 0.29  | 0.29  | 0.52  | 0.53  | 0.46  | 0.55  | 0.41  | 0.42  | 0.33  | 0.26  | 0.15  | (45) |  |

## Historical Trends

|                              |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                          | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) Station Only            | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (13 neighbors) | N/A | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page